

Raymond “Ray” Hernandez Parcon, M.Sc.

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RESEARCH & TEACHING INTERESTS

- Host-Pathogen Interaction and Plant Disease Resistance
- Molecular Genetics and Genomics
- Active, Blended, and Experiential Learning Pedagogies

EDUCATION

- 202x **Doctor of Philosophy in Plant Pathology**
Department of Plant Pathology, [University of Georgia](#)
Advisors: Dr. Shavannor Smith and Dr. James Buck
Proposed dissertation: “Characterization of *Cercospora sojina*’s Genetic Diversity to Better Understand the Resistance Mechanisms of Frogeye Leaf Spot Disease in Soybean.”
- 2021 **Master of Science in Microbiology**
Institute of Biology, [University of the Philippines Diliman](#)
Advisor: Dr. Jessica Simbahan
Thesis: “Efficacy of a Microbial Biological Control Agent (MBCA) in Controlling the Growth of Fungal Pathogens of Tomato
- 2014 **Bachelor of Science in Biology**
Biology Department, [Adamson University](#)
Advisor: Ace Bryan Cabal, M.Sc.
Thesis: “Diversity of Streptomyces in Las Piñas-Parañaque Critical Habitat and Ecotourism Area, Metro Manila, Philippines: A Pilot Study.”

RESEARCH EXPERIENCE

Research

- 202x **Graduate Research Assistant (ongoing)**
Department of Plant Pathology, University of Georgia, GA, USA
- Studies the genetic diversity of *Cercospora sojina*, the causal agent of frogeye leaf spot in soybeans by identifying haplotypes across spatial and temporal collections of the pathogen using various CLI tools and software (e.g., Git Bash, R, STRUCTURE, Geneious, etc.).
 - Developed bioinformatic pipelines for SNP discovery from genotyping-by-sequencing data and *de novo* genome assembly using high-performance computing.
- 2021 **Project Consultant (3 mo.)**
Institute of Biology, University of the Philippines Diliman, Quezon City, Philippines
- Research consultant in the production and improvement of construction materials using bacterial calcite precipitation.
- 2020 **University Research Associate 2 (2 yr. 5 mo.)**
Institute of Biology, University of the Philippines Diliman, Quezon City, Philippines

- Conducted research using a polyphasic approach: integrating morphological, physiological, and molecular analyses of a biologically active microorganism, and analyzed qualitative and quantitative data of its inhibitory effects against fungal pathogens of tomato. Additionally, managed administrative tasks related to the project.
- 2018 **University Research Associate 1** (1 yr. 5 mo.)
Institute of Biology, University of the Philippines Diliman, Quezon City, Philippines
- Conducted research on biologically active microorganisms by screening them through various metabolic assays and analyzing qualitative and quantitative physiological data of a potential biologically active enzyme against a collection of fungal plant pathogens. Additionally, managed administrative tasks related to the project.

Relevant Trainings & Workshops

- 2022 Software Carpentry Core Curricula (Unix Shell, Git, and Python) Workshop, Carpentries@UGA
- 2022 Data Carpentry Genomics Curriculum Workshop, Carpentries@UGA

TEACHING & MENTORSHIP EXPERIENCE

Teaching Experience

- 2023 **Teaching Assistant**
Department of Plant Pathology, University of Georgia, Athens, GA, USA
- Assisted Dr. Harald Scherm, Dr. Melissa Mitchum, and Dr. Shavannor Smith in designing experiments, preparing laboratory materials, and shared grading responsibilities for Introductory Plant Pathology for Spring and Fall semesters.
 - Led laboratory sessions for students and supported active student learning.
- 2021 **Instructor 3**
Department of Biochemistry, University of Santo Tomas, Manila City, Philippines
- Instructor of record for Basic Inorganic and Organic Chemistry laboratory course for First semester (Fall semester).
 - Applied active and blended learning strategies to enhance student engagement in an online learning environment.
 - Faculty Competence Evaluation Score: 22/25 or Very Satisfactory (4-point rating scale)
- 2015 **Instructor 1**
Biology Department, Adamson University, Manila, Philippines
- Instructor of record for Human Biology, Biological Science, Zoology (laboratory), and Human Anatomy and Physiology courses for First semester (Fall semester).
 - Developed course activities and assessments grounded in active and experiential learning pedagogies.
 - Teacher's Behavior Inventory Rating: 95% or Excellent (5-point rating scale)

Mentorship

- 202x Worked closely with two mentees through GradMap Philippines to develop individualized mentorship plans for their international STEM graduate school applications.
- 2023 Mentored a high school intern through the UGA CAES Young Scholars Internship Program, helping them earn first place in the Athens campus poster competition.

Invited Lectures

- 2025 *Institute of Biology, University of the Philippines Diliman, Quezon City, Philippines*

- Developed presentation slides and delivered a lecture on “Foliar Plant Pathogens in Soil” in Dr. Jessica Simbahan’s Soil Microbiology 3-unit graduate course.
- 2022 *Institute of Biology, University of the Philippines Diliman, Quezon City, Philippines*
- Developed presentation slides and delivered a lecture on “Perspectives on Plant-Microbe Interactions” in Dr. Marie Christine Obusan’s Advanced Microbial Ecology 3-unit graduate course.

Pedagogical Trainings & Workshops

- 202x GradTeach Certificate, Center for Teaching and Learning, University of Georgia
- 2026 Certified Instructor, Data, Library, and Software Carpentries, The Carpentries
- 2026 Active Learning Summit 2026, University of Georgia

PUBLICATIONS

Peer-reviewed Journal Articles

1. **Parcon RH**, Cruz RJB, Rey JD, Dacones LS, Simbahan JF. (2026). *Bacillus amyloliquefaciens* N2B2 and its potential for co-cultivation with chitinolytic bacteria to control the growth of *Agroathelia rolfsii* and other fungal pathogens of tomatoes (*Solanum lycopersicum* L.). *Journal of Biological Control*, 40(1):1-15. <https://doi.org/10.18311/jbc/2026/49421>
2. Supnet SJG, Caballero EVM, **Parcon RH**, Simbahan JF. (2020). Marilao-Meycauayan-Obando River System (MMORS) Harbors Multidrug-Resistant Bacteria Indicating High Risk of Antimicrobial Contamination. *Philippine Science Letters*, 13 (Supplement); 198-205. URL: <https://www.philsciletters.net/2020-198/>
3. Atole LDC, Isagan MDE, **Parcon RH**, Alcantara VA, Zarate JT, Simbahan JF. (2020). Variations in enzymatic activity and response to abiotic factors of endophytic bacteria isolated from Narra (*Pterocarpus indicus*) growing in a forest environment and urban arboretum. *Philippine Science Letters*, 13 (Supplement); 120-126. URL: <https://www.philsciletters.net/2020-special-issue-12/>
4. **Parcon RH**, Isagan MDE, Atole LDC, Alcantara VA, Simbahan JF. *Bacillus amyloliquefaciens*, a Narra (*Pterocarpus indicus*) leaf endophyte, possesses antifungal activity against phytopathogenic fungi due to laminarinase activity and production of antimicrobial compounds. (2019). *Philippine Science Letters*, 12 (Supplement); 048-054. URL: <https://philsciletters.net/2019-special-issue-4/>

Contributed Talks

- 2026 “Tracking 44 Years of Frogeye Leaf Spot: Predominant *Cercospora soja* Genotypes as Priority Targets for Host Resistance.” 2026 Annual Meeting of the Georgia Association of Plant Pathologists, Jekyll Island, GA
- 2026 “Characterization of the genetic diversity and virulence of historical and current isolates of *Cercospora soja* collected over 42 years.” 103rd Annual Meeting of the Southern Division of the American Phytopathological Society, Athens, GA

Poster Presentations at Professional Meetings

- 2026 Plant Health 2026, American Phytopathological Society, Providence, RI
- 2025 Plant Health 2025, American Phytopathological Society, Honolulu, HI
- 2025 19th Biennial Conference on Molecular & Cellular Biology of the Soybean, Madison, WI

AWARDS & RECOGNITIONS

- 2026 Finalist, Three-Minute Thesis (3MT®) – PhD Category, College of Agricultural and Environmental Sciences, University of Georgia
- 2026 Domestic Travel Grant, Graduate School, University of Georgia
- 2026 1st Place, Oral Paper Presentation – Annual Meeting of Georgia Association of Plant Pathologists
- 2026 1st Place, Three-Minute Thesis (3MT®) – PhD Category, Department of Plant Pathology, University of Georgia
- 2025 Finalist Alternate, Three-Minute Thesis (3MT®), Graduate School, University of Georgia
- 2025 International Travel Grant, Graduate School, University of Georgia
- 2025 Summer Research Grant, Graduate School, University of Georgia
- 2024 Outstanding Teaching Assistant Award, Center for Teaching and Learning, University of Georgia

LEADERSHIP & SERVICE

- 2026 Volunteer Mentor, GradMap Philippines Cycle 26-27
- 2026 Volunteer, APS Gives Back: Providence Parks, Providence, RI
- 2026 Moderator, Graduate Student Research Competition Session, 103rd Annual Meeting of the Southern Division of the American Phytopathological Society, Athens, GA
- 2025 Volunteer, APS Gives Back: Native Roots Experience, Honolulu, HI
- 2025 Chapter Reviewer, Mycology in a Changing Planet: Applications and Perspectives from the Southeast Asia (Springer Nature, 2025)
- 2023 Secretary, Society of Aspiring Plant Pathologists, University of Georgia (FY 2022-23)

PROFESSIONAL & HONOR SOCIETY MEMBERSHIPS

1. American Phytopathological Society
2. American Phytopathological Society - Southern Division
3. Society of Aspiring Plant Pathologists, University of Georgia
4. The Honor Society of Phi Kappa Phi, University of the Philippines Diliman

REFERENCES
